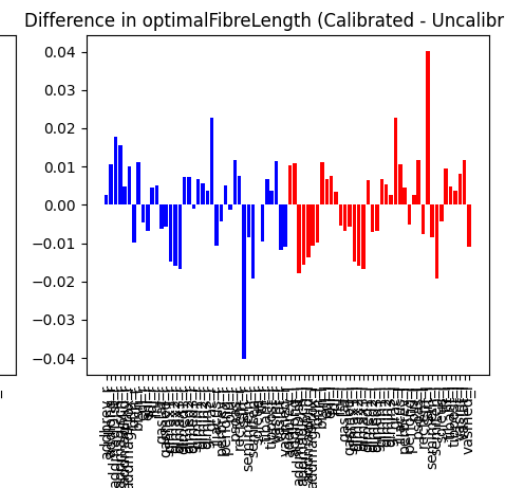
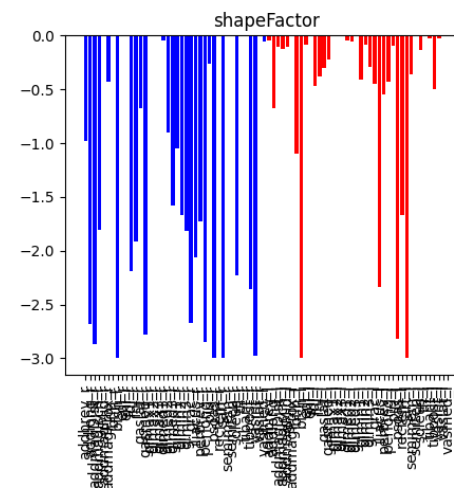
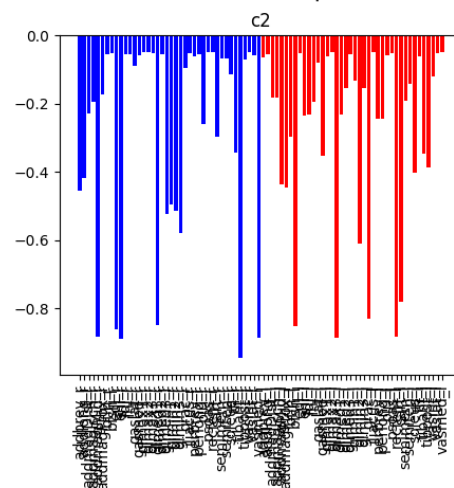
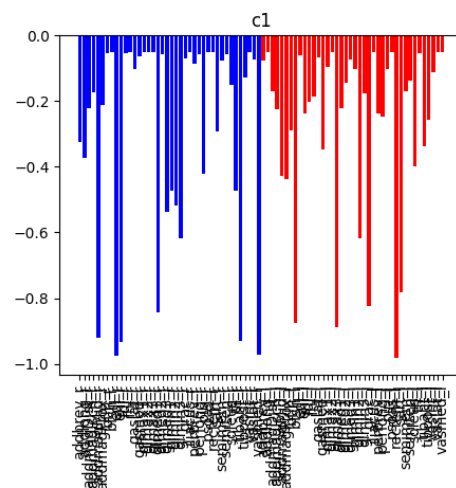




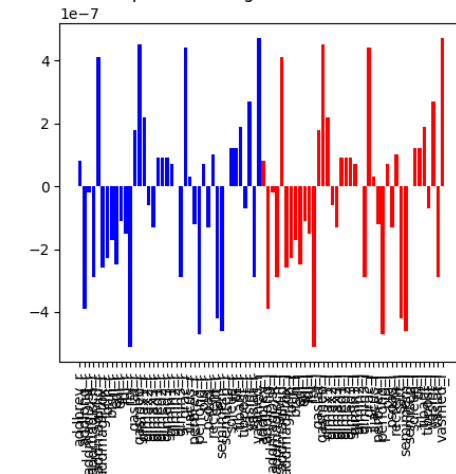
Scaled_model

CEINMS
Calibration

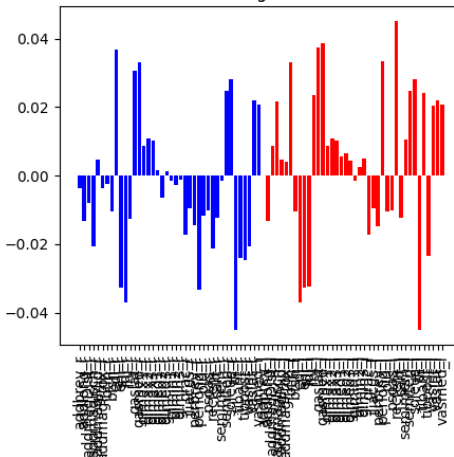
Compare: uncalibrated vs calibrated



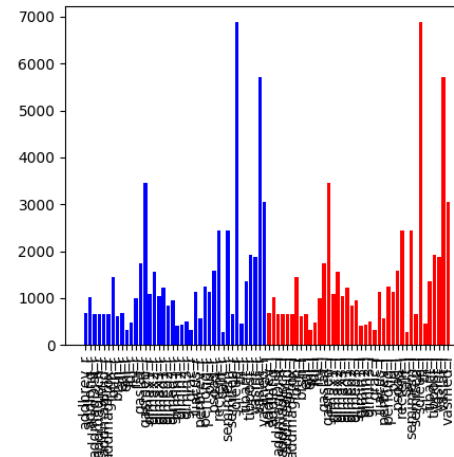
Difference in pennationAngle (Calibrated - Uncalibrated)



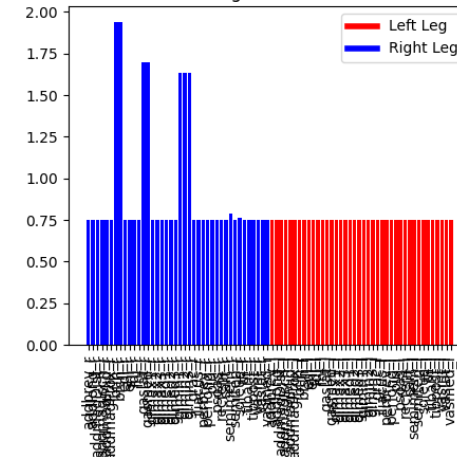
Difference in tendonSlackLength (Calibrated - Uncalibrated)



maxIsometricForce



strengthCoefficient



Scaled_model

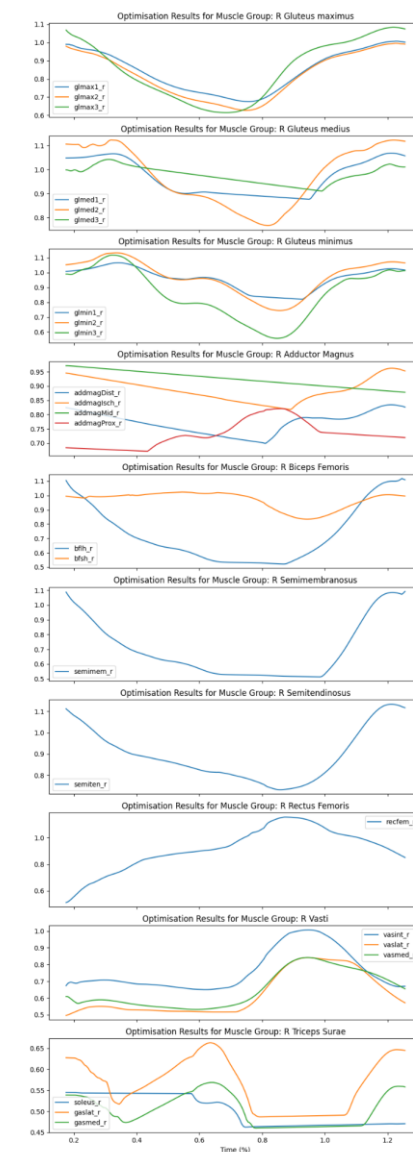
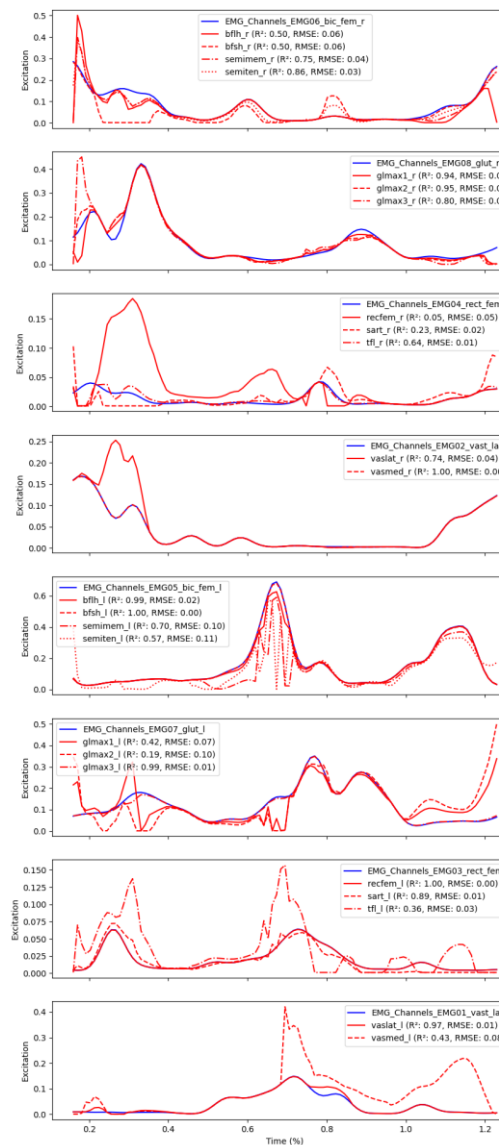
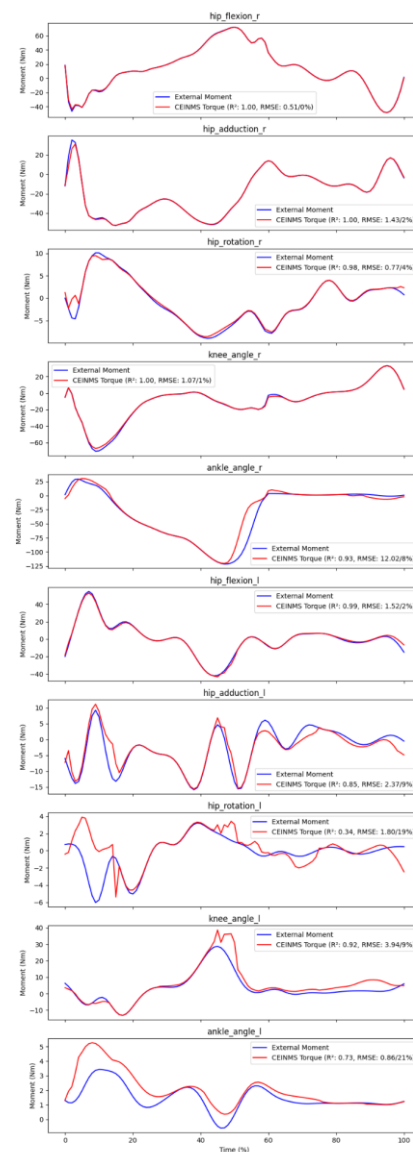
CEINMS
Calibration

Optimise

Alpha = 1

Beta = 1

Gamma = 187

Moments error better < 20%
for all



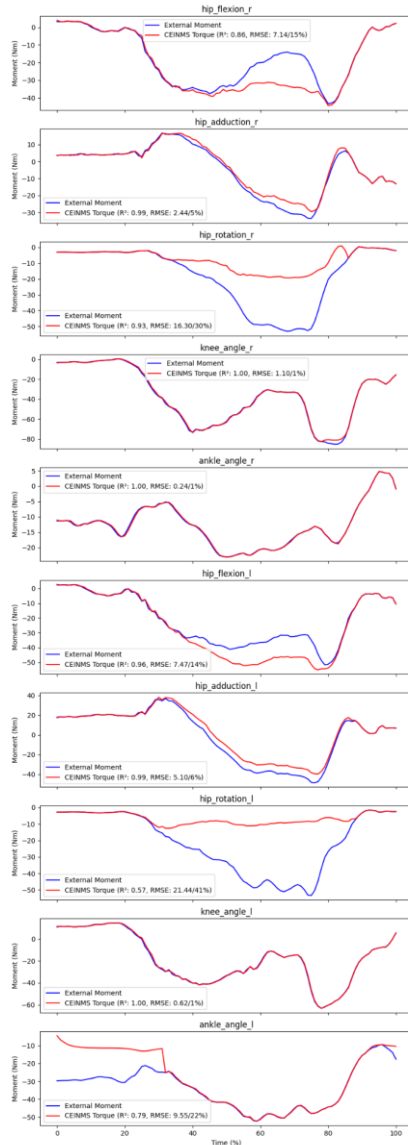
Scaled_model

CEINMS
Calibration

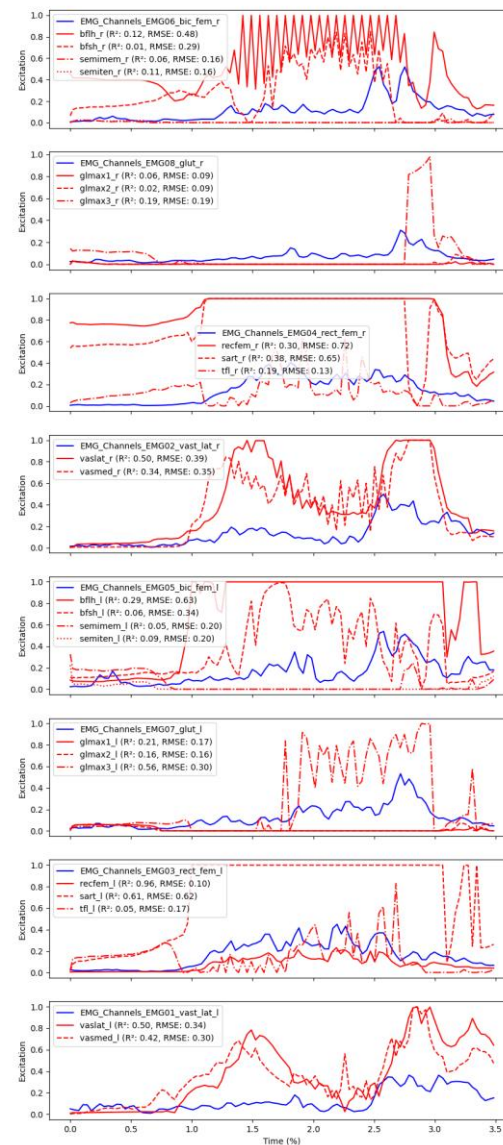
Exe
Alpha = 10000

Model not able to produce
moments for hip rotation even
with very high Alpha

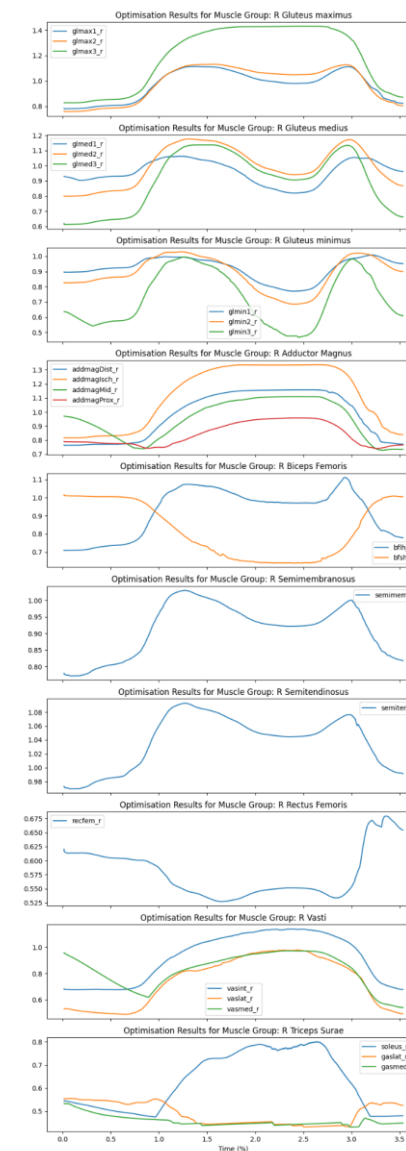
External Torques vs CEINMS Torques



EMG vs CEINMS Excitations



Optimisation Results: NormFibreLengths.sto

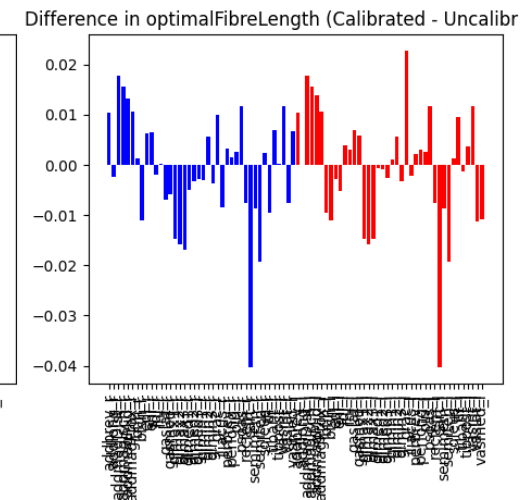
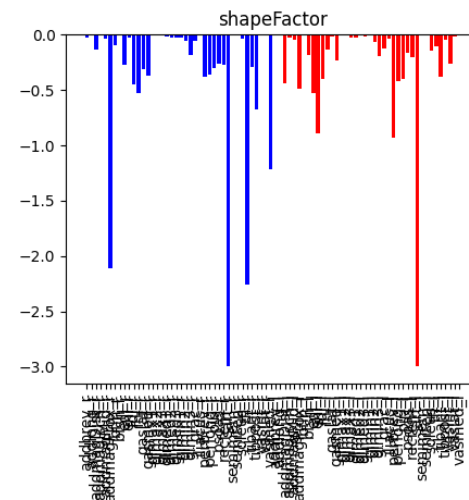
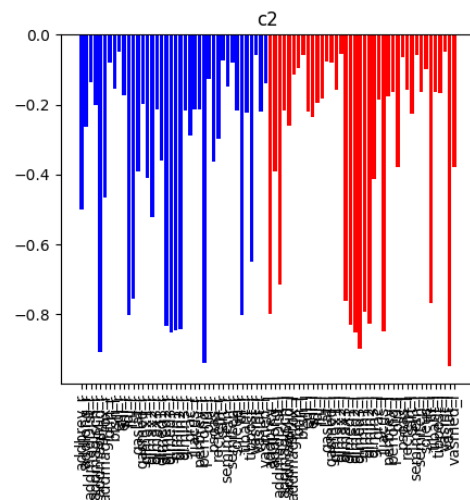
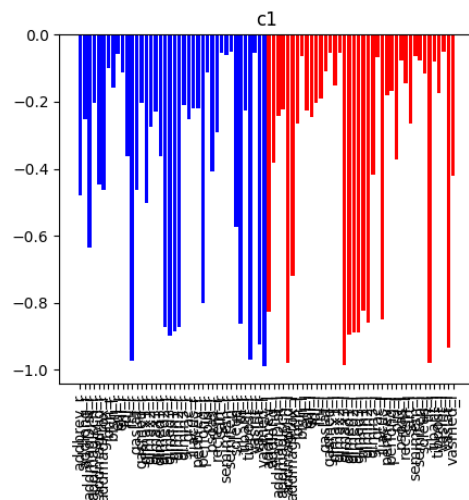




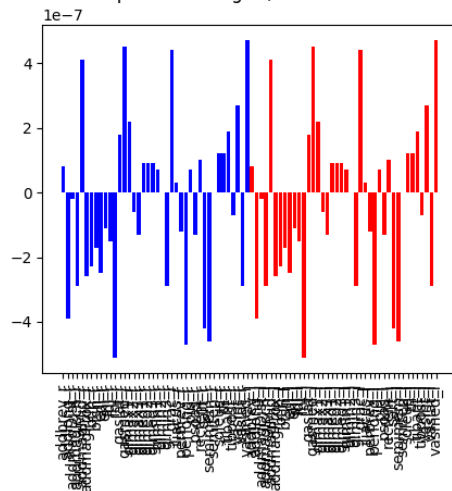
Scaled_model

CEINMS
Calibration

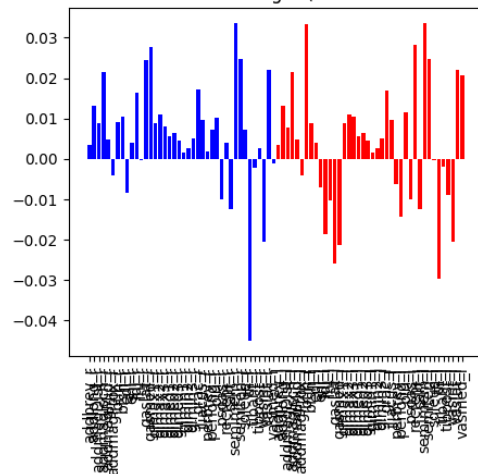
Compare: uncalibrated vs calibrated



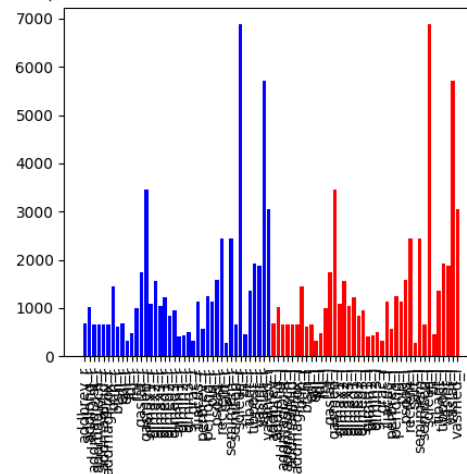
Difference in pennationAngle (Calibrated - Uncalibrated)



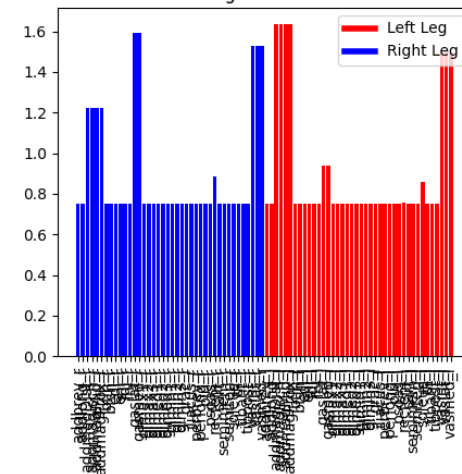
Difference in tendonSlackLength (Calibrated - Uncalibrated)



maxIsometricForce



strengthCoefficient



Compared with walking calibration, OFL is not as increased (max diff = 0.04 vs 0.02m)



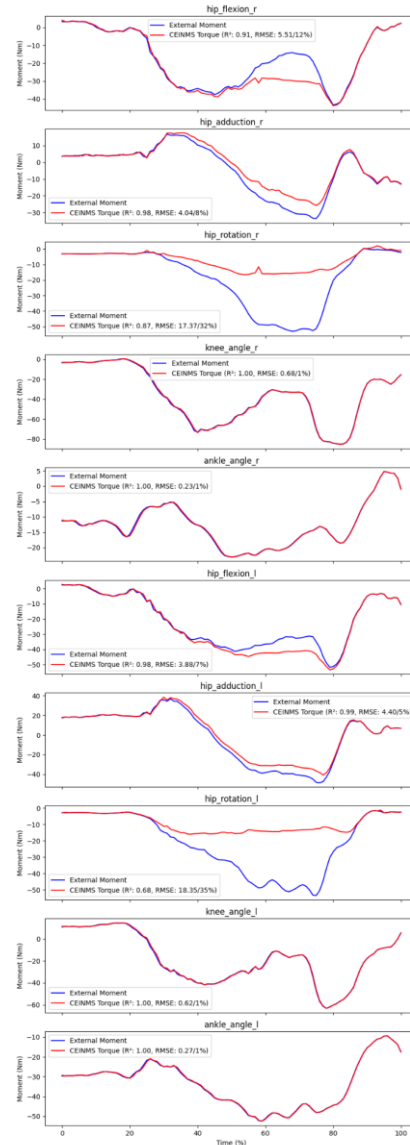
Scaled_model

CEINMS
Calibration

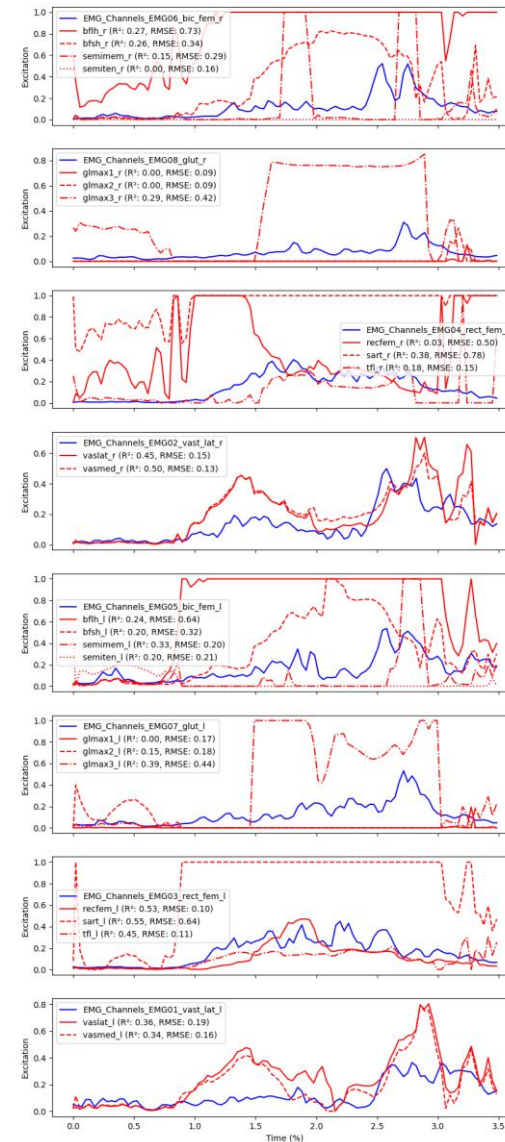
Exe
Alpha = 10000

Model not able to produce
moments for hip rotation even
with very high Alpha

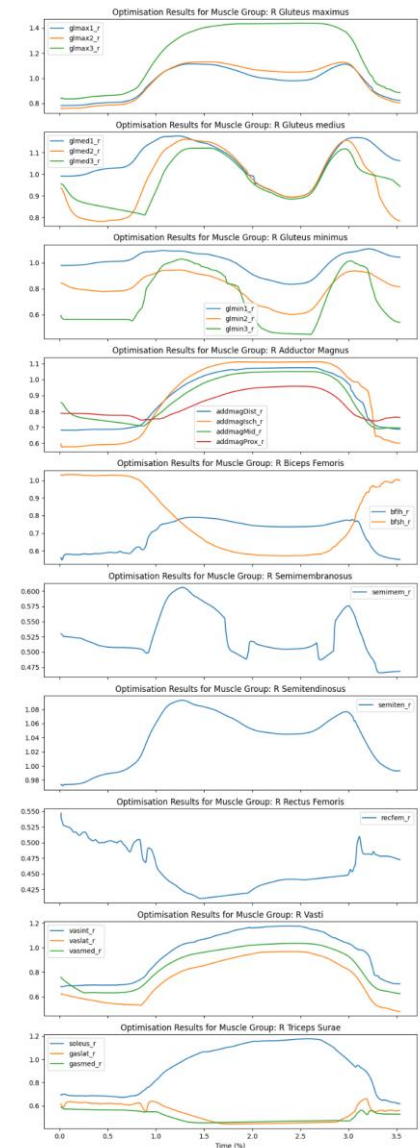
External Torques vs CEINMS Torques



EMG vs CEINMS Excitations



Optimisation Results: NormFibreLengths.sto





Scaled_model

CEINMS
Calibration

Optimal

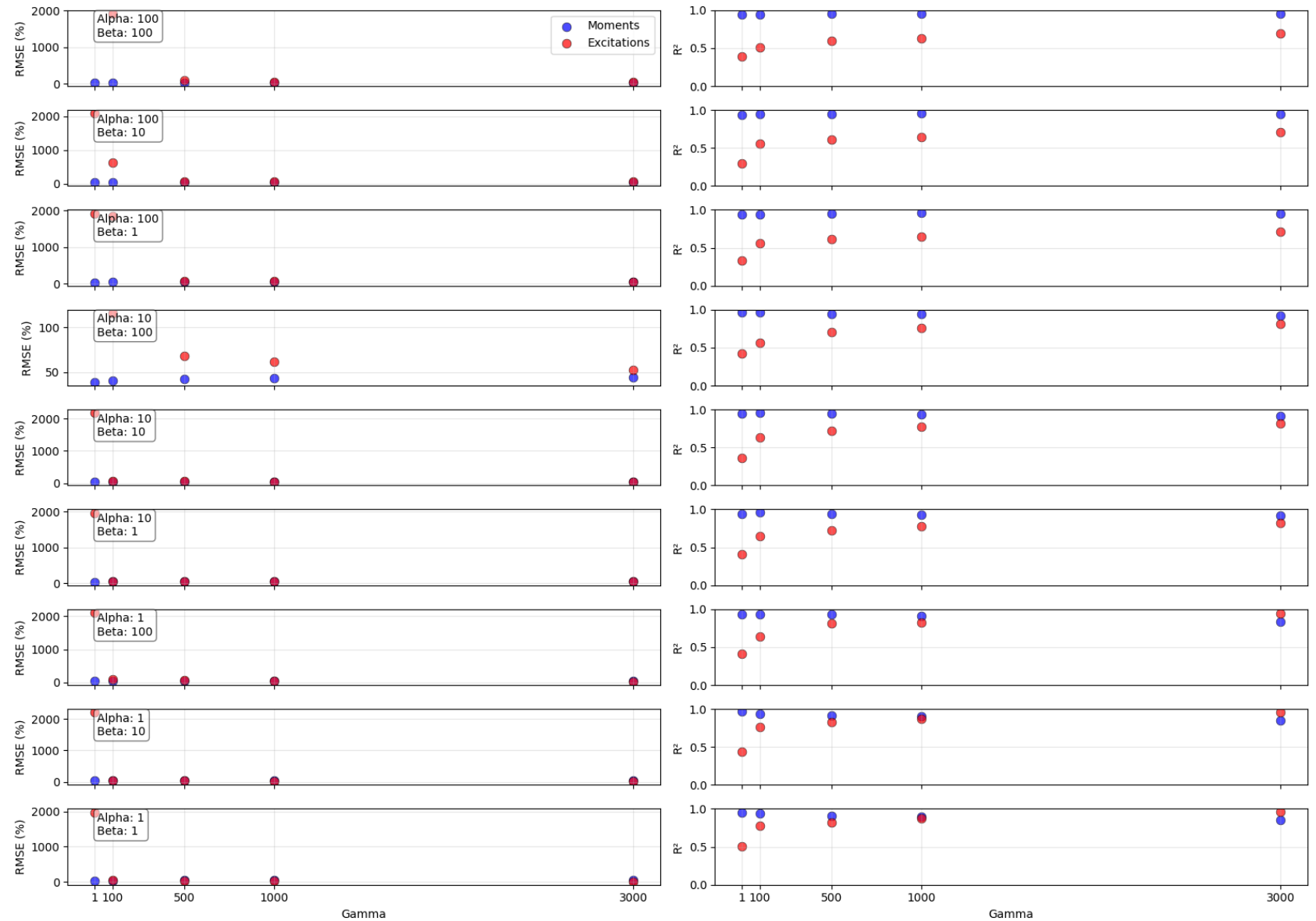
A = 1

B = 1

G = 3000

-

Best Overall - Alpha: 1, Beta: 1, Gamma: 3000
Combined RMSE: 68.44, Combined R²: 1.81



Scaled_model

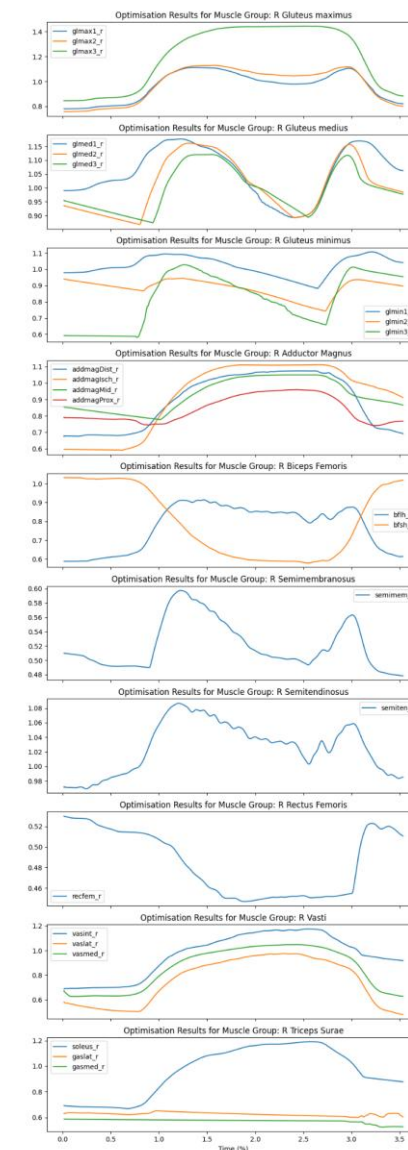
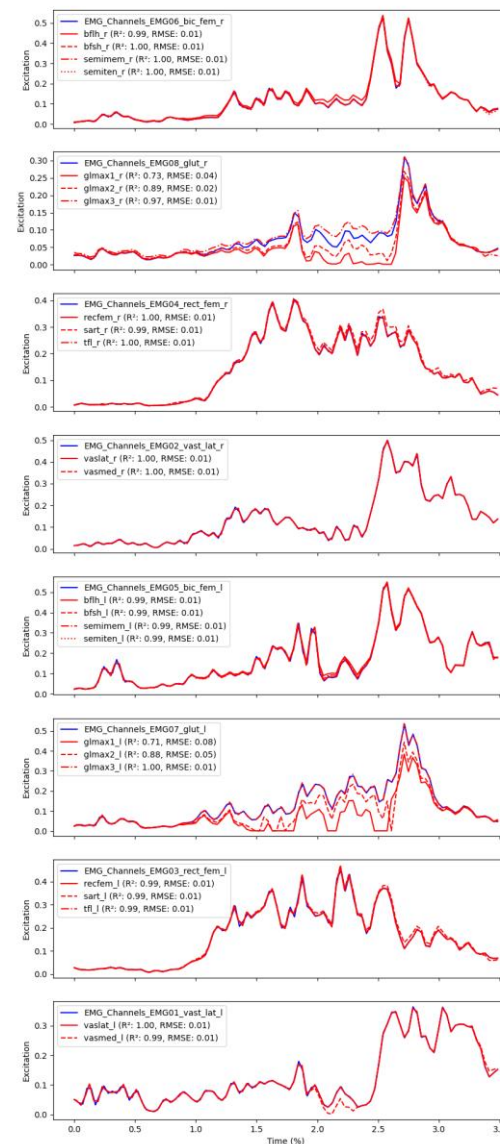
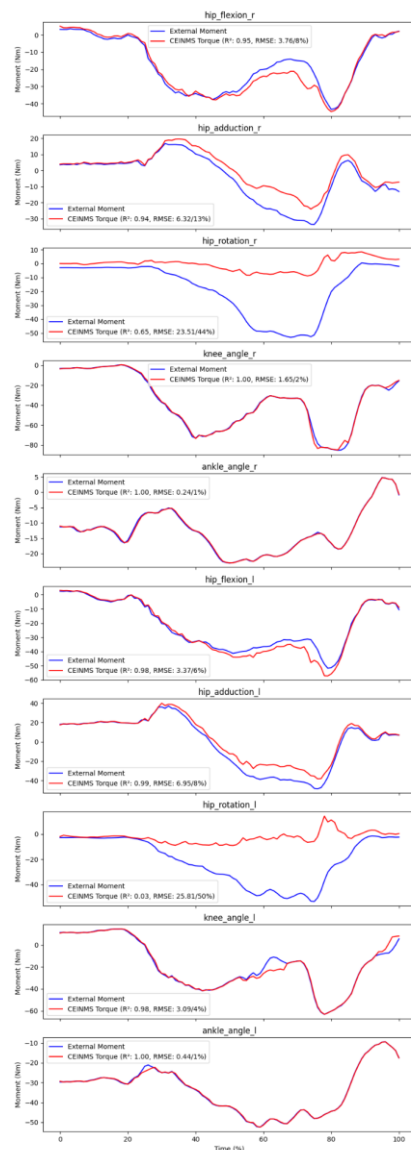
CEINMS
Calibration

Optimal

A = 1

B = 1

G = 3000

Moooment and emg tracking
good except Hip rotationSemimemb and RectFem with
very low norm fibre lengths
(~0.5)

Scaled_model

CEINMS
Calibration

Execution

Alpha = 10

Beta = 1

Gamma = 187

Walking with squat
calibration seems to be
good and similar to
walking calibration

